Given a string w and a symbol a, let $delete(a, w)$ be the string w with all instances of a removed. For example, $delete(\texttt{z}, \texttt{jazzy})$ is `jay` and $delete(\texttt{1}, \texttt{00101110})$ is `0000`.

(a) Give a recursive function to compute *delete*.

Solution: Let a be a symbol and w a string. (Hint: there are at least two cases based on the recursive definition of strings, but you may need more than two!)

$$delete(a, w) := \begin{cases} & \text{if} \\ & \text{if} \end{cases}$$

■

Collaborators and sources:

- Bob Smith helped me. He took this course at UM in Fall 2020. He suggested that I needed two separate recursive cases.
- I looked at the slides at https://chengjianglong.com/slides/DS_Lecture_22.pdf for another perspective on recursive structures.
- I used ChatGPT. Here is the full transcript of my conversation: https://chatgpt.com/share/6970e55f-e594-800e-88e3-c5eaa8caf7ce

(b) Prove that $delete(a, x \bullet y) = delete(a, x) \bullet delete(a, y)$ for all strings x, y and all symbols a.

Solution: Let x, y be arbitrary strings and a an arbitrary symbol.

Assume... (what's the inductive hypothesis?)

There are ?? cases to consider:

- If $w = \varepsilon$, then

$$
\begin{aligned}
delete(a, x \bullet y) &= delete(a, \varepsilon \bullet y) && \text{because ?} \\
&= delete(a, \varepsilon) \bullet delete(a, y) && \text{because ?} \\
&= delete(a, x) \bullet delete(a, y) && \text{because } x = \varepsilon
\end{aligned}
$$

- ...you fill in the rest of the cases

In all cases, we conclude that $delete(a, x \bullet y) = delete(a, x) \bullet delete(a, y)$. ■

Collaborators and sources:

- Same ChatGPT conversation as above. Here is the full transcript of my conversation: https://chatgpt.com/share/6970e55f-e594-800e-88e3-c5eaa8caf7ce

(c) Recall the function $\#(a, w)$ from problem session 1, which returns the number of occurrences of the symbol a in string w. For any string $w \in \{0,1\}^*$, prove that $|delete(1, w)| = \#(0, w)$.

Solution: Let $w \in \{0,1\}^*$.

In all cases, we conclude that $|delete(1, w)| = \#(0, w)$. ■

Collaborators and sources: None.